WSPÓŁRZĘDNE KRZYWOLINIOWE (q_1, q_2, q_3)

ukł. kartezjański:

$$\vec{r} = \hat{i}x + \hat{j}y + \hat{k}z \qquad (0)$$

teraz:

$$q_1 = q_1(x,y,z) \qquad (1a)$$
$$q_2 = q_2(x,y,z) \qquad (1b)$$
$$q_3 = q_3(x,y,z) \qquad (1c)$$

Współczynniki Lamego:

$$\boxed{H_l = \left| \frac{\partial \vec{r}}{\partial q_l} \right|} \qquad (2) \quad , \; l \in \{1,2,3\}$$

Wersory:

$$\boxed{\hat{q}_l = \frac{1}{H_l} \frac{\partial \vec{r}}{\partial q_l}} \qquad (3)$$

$$\begin{bmatrix} \hat{q}_1 \\ \hat{q}_2 \\ \hat{q}_3 \end{bmatrix} = A \begin{bmatrix} \hat{i} \\ \hat{j} \\ \hat{k} \end{bmatrix} \quad (4a) \text{, zatem} \quad \begin{bmatrix} \hat{i} \\ \hat{j} \\ \hat{k} \end{bmatrix} = A^{-1} \begin{bmatrix} \hat{q}_1 \\ \hat{q}_2 \\ \hat{q}_3 \end{bmatrix} \quad (4b)$$

macierz 3×3

$= A^T$, bo $\hat{q}_1, \hat{q}_2, \hat{q}_3$ są ortonormalne (*)

Dowolny wektor:

$$\vec{w} = w_1 \hat{q}_1 + w_2 \hat{q}_2 + w_3 \hat{q}_3 = \qquad (5)$$
$$= w_x \hat{i} + w_y \hat{j} + w_z \hat{z}$$

Niech $f \to$ pewna funkcja skalarna

- gradient f, grad(f)

$$\boxed{\nabla f = \sum_{l=1}^{3} \frac{1}{H_l} \left(\frac{\partial f}{\partial q_l} \right) \hat{q}_l} \qquad (6)$$

- dywergencja $\vec{w}$, div$(\vec{w})$

$$\boxed{\nabla \cdot \vec{w} = \frac{1}{H_1 H_2 H_3} \left(\frac{\partial (w_1 H_2 H_3)}{\partial q_1} + \frac{\partial (H_1 w_2 H_3)}{\partial q_2} + \frac{\partial (H_1 H_2 w_3)}{\partial q_3} \right)} \qquad (7)$$

- rotacja $\vec{w}$, $rot(\vec{w})$

$$\boxed{\nabla \times \vec{w} = \frac{1}{H_1 H_2 H_3} \begin{vmatrix} H_1 \hat{q}_1 & H_2 \hat{q}_2 & H_3 \hat{q}_3 \\ \frac{\partial}{\partial q_1} & \frac{\partial}{\partial q_2} & \frac{\partial}{\partial q_3} \\ H_1 w_1 & H_2 w_2 & H_3 w_3 \end{vmatrix}} \qquad (8)$$

- laplasjan

→ skalarny: $\Delta \varphi = div(grad(\varphi))$

$$\boxed{\Delta \varphi = \frac{1}{H_1 H_2 H_3} \left[\frac{\partial}{\partial q_1} \left(\frac{H_2 H_3}{H_1} \frac{\partial \varphi}{\partial q_1} \right) + \frac{\partial}{\partial q_2} \left(\frac{H_1 H_3}{H_2} \frac{\partial \varphi}{\partial q_2} \right) + \frac{\partial}{\partial q_3} \left(\frac{H_1 H_2}{H_3} \frac{\partial \varphi}{\partial q_3} \right) \right]} \qquad (9a)$$

→ wektorowy: $\Delta \vec{w} = ?$

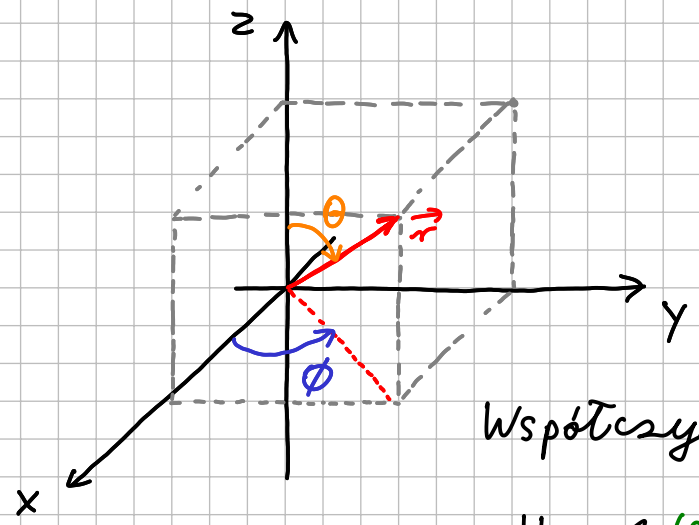
ukł. kartezjański $\Delta \vec{w} = \hat{i} \Delta w_x + \hat{j} \Delta w_y + \hat{k} \Delta w_z$ (9b')

Jednak w ogólności (9b') NIE DZIAŁA dla współrzędnych krzywoliniowych !

mamy w ogólności, że

$$\boxed{\Delta \vec{w} = \underset{grad}{\nabla} \left(\underset{div}{\nabla \cdot} \vec{w} \right) - \underset{rot}{\nabla} \times \left(\underset{rot}{\nabla \times} \vec{w} \right)} \qquad (9b)$$

PRZYKŁAD - UKŁAD SFERYCZNY (MATEMATYCZNY)



$$\begin{cases} x = r\cos\phi\sin\theta & (1a) \\ y = r\sin\phi\sin\theta & (1b) \\ z = r\cos\theta & (1c) \end{cases}$$

Współczynniki Lamego:

$$H_r = 1 \ (2a), \quad H_\theta = r \ (2b), \quad H_\phi = r\sin\theta \ (2c)$$

Wersory:

$$\begin{cases} \hat{r} = [\cos\phi\sin\theta, \sin\phi\sin\theta, \cos\theta] & (3a) \\ \hat{\theta} = [\cos\phi\cos\theta, \sin\phi\cos\theta, -\sin\theta] & (3b) \\ \hat{\phi} = [-\sin\phi, \cos\phi, 0] & (3c) \end{cases}$$

zatem:

$$\begin{bmatrix} \hat{r} \\ \hat{\theta} \\ \hat{\phi} \end{bmatrix} = \begin{bmatrix} \cos\phi\sin\theta & \sin\phi\sin\theta & \cos\theta \\ \cos\phi\cos\theta & \sin\phi\cos\theta & -\sin\theta \\ -\sin\phi & \cos\phi & 0 \end{bmatrix} \begin{bmatrix} \hat{i} \\ \hat{j} \\ \hat{k} \end{bmatrix} \qquad (4a)$$

$\downarrow$ transpozycja

$$\begin{bmatrix} \hat{i} \\ \hat{j} \\ \hat{k} \end{bmatrix} \begin{bmatrix} \cos\phi\sin\theta & \cos\phi\cos\theta & -\sin\phi \\ \sin\phi\sin\theta & \sin\phi\cos\theta & \cos\phi \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \begin{bmatrix} \hat{r} \\ \hat{\theta} \\ \hat{\phi} \end{bmatrix} \qquad (4b)$$

stąd

$$\begin{cases} \hat{i} = \hat{r}\cos\phi\sin\theta + \hat{\theta}\cos\phi\cos\theta - \hat{\phi}\sin\phi & (3a') \\ \hat{j} = \hat{r}\sin\phi\sin\theta + \hat{\theta}\sin\phi\cos\theta + \hat{\phi}\cos\phi & (3b') \\ \hat{k} = \hat{r}\cos\theta - \hat{\theta}\sin\theta & (3c') \end{cases}$$

Niech teraz $\vec{w} = w_r\hat{r} + w_\theta\hat{\theta} + w_\phi\hat{\phi}$ (5)

- gradient f

$$\boxed{\nabla f \overset{(2),(6)}{=} \left(\frac{\partial f}{\partial r}\right)\hat{r} + \frac{1}{r}\left(\frac{\partial f}{\partial \theta}\right)\hat{\theta} + \frac{1}{r\sin\theta}\left(\frac{\partial f}{\partial \phi}\right)\hat{\phi}} \qquad (6)$$

- dywergencja $\vec{w}$

$$\mathrm{div}(\vec{w}) \overset{(2),(7)}{=} \frac{1}{r^2\sin\theta}\left(\frac{\partial(w_r r^2\sin\theta)}{\partial r} + \frac{\partial(w_\theta r\sin\theta)}{\partial\theta} + \frac{\partial(r w_\phi)}{\partial\phi}\right)$$

$$\boxed{\mathrm{div}(\vec{w}) = \frac{1}{r^2}\frac{\partial(r^2 w_r)}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial(w_\theta\sin\theta)}{\partial\theta} + \frac{1}{r\sin\theta}\frac{\partial w_\phi}{\partial\phi}} \qquad (7)$$

- rotacja

$$\mathrm{rot}(\vec{w}) \overset{(2),(8)}{=} \frac{1}{r^2\sin\theta}\begin{vmatrix} \hat{r} & r\hat{\theta} & r(\sin\theta)\hat{\phi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial\theta} & \frac{\partial}{\partial\phi} \\ w_r & r w_\theta & r(\sin\theta) w_\phi \end{vmatrix} =$$

$$= \frac{1}{r^2\sin\theta}\left(\hat{r}\frac{\partial(r(\sin\theta)w_\phi)}{\partial\theta} - \hat{r}\frac{\partial(r w_\theta)}{\partial\phi} - r\hat{\theta}\frac{\partial(r(\sin\theta)w_\phi)}{\partial r} + r\hat{\theta}\frac{\partial w_r}{\partial\phi} + r(\sin\theta)\hat{\phi}\frac{\partial(r w_\phi)}{\partial r} - r(\sin\theta)\hat{\phi}\frac{\partial w_r}{\partial\theta}\right) =$$

$$= \frac{\cancel{r}\,\hat{r}}{r^{\cancel{2}}\sin\theta}\left(\frac{\partial(w_\phi\sin\theta)}{\partial\theta} - \frac{\partial w_\theta}{\partial\phi}\right) - \frac{\cancel{r}\,\hat{\theta}}{r^{\cancel{2}}\sin\theta}\left(\sin\theta\frac{\partial(r w_\phi)}{\partial r} - \frac{\partial w_r}{\partial\phi}\right) + \frac{\cancel{r}\,\hat{\phi}\,\cancel{\sin\theta}}{r^{\cancel{2}}\cancel{\sin\theta}}\left(\frac{\partial(r w_\phi)}{\partial r} - \frac{\partial w_r}{\partial\theta}\right)$$

$$\boxed{\mathrm{rot}(\vec{w}) = \frac{\hat{r}}{r\sin\theta}\left(\frac{\partial(w_\phi\sin\theta)}{\partial\theta} - \frac{\partial w_\theta}{\partial\phi}\right) + \frac{\hat{\theta}}{r}\left(\frac{1}{\sin\theta}\frac{\partial w_r}{\partial\phi} - \frac{\partial(r w_\phi)}{\partial r}\right) + \frac{\hat{\phi}}{r}\left(\frac{\partial(r w_\phi)}{\partial r} - \frac{\partial w_r}{\partial\theta}\right)} \qquad (8)$$

- Laplasjan (skalarny)

$$\Delta f \overset{(2),(9a)}{=} \frac{1}{r^2 \sin\theta}\left[\frac{\partial}{\partial r}\left(r^2 \sin\theta \frac{\partial f}{\partial r}\right) + \frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial f}{\partial \theta}\right) + \frac{\partial}{\partial \phi}\left(\frac{1}{\sin\theta}\frac{\partial f}{\partial \phi}\right)\right]$$

$$\boxed{\Delta f = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial f}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial f}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2 f}{\partial \phi^2}} \qquad (9)$$